Short-term Forecasting of Solar Radiation Using a Hybrid Model of CNN-LSTM Integrated with EEMD

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***Abstract*— Nowadays, renewable energy gradually become indispensable sources all over the world such as solar energy, wind energy, tidal energy, etc. In terms of solar energy, solar radiation fluctuates and depends on various other factors. Therefore, short-term forecasting of solar radiation plays a consistently important role in many fields of solar energy applications, especially in generating electricity. In this paper, we proposed a Convolutional Neural Network – Long-Short Term Memory (CNN-LSTM) network integrated with the Ensemble Empirical Mode Decomposition (EEMD) method to make a short-term forecast of solar irradiation in Vietnam. Firstly, we used EEMD method to separate the original irradiation series into intrinsic mode functions (IMFs). Secondly, each IMFs were fed into a predictive model that combined CNN and LSTM network and then composed into final forecasting of the solar irradiation. Finally, the results were compared with that of other methods such as the single model of CNN, LSTM and Bi-directional-LSTM to find out the benefits. The comparison illustrated that the performance of the proposed model was better than the others, namely the n-RMSE was 0.098 while that of LSTM, Bi-LSTM and CNN was 0.187, 0.169 and 0.177, respectively.**

Keywords*— Short-term forecasting, solar radiation, hybrid model, EEMD, CNN-LSTM.

I. INTRODUCTION

While the global energy demand increases, solar power is becoming an indispensable power in the world because of the reducing carbon emission and is alternative energy for fossil fuels, especially in the electricity generating fields [1]. In 2016, the installed capacity of renewable energy in Vietnam accounted for only about 0.4% of the total installed capacity of the country equivalent to 135 megawatts (MW), while in 2020 solar power accounted for 12.8% and rooftop solar power was recorded at 11.23% of total supply has shown the rapid development of renewable energy sources, with 8,871 MW and 7,785 MW of installed capacity, respectively [2], [3]. However, the fluctuation of solar irradiation caused by short-term events such as suddenly shading of moving clouds can impact the PV power and the stability of smart grids. Therefore, solar radiation forecasting (SRF) plays an important role in increasing the stability of distributed grids and improving the quality of electricity supplied to consumers.

There have been many studies in the SRF fields [4]. The predictive techniques can be roughly classified into five categories: Physical, statistical, machine learning methods, deep learning methods and hybrid methods. Physical methods, for example, numerical weather prediction (NWP) [5], require solar irradiance and numerous meteorological parameters to develop a complex physical mathematical model. Nevertheless, for those with incomplete or unavailable information in the plant design documentation such as PV plant owners, this approach is not suitable, and it is also complicated to calculate. Statistical models, such as the autoregressive (AR) model, autoregressive integrated moving average (ARIMA) [6] or exponential smoothing [7] are usually used to forecast short-term solar radiation [8].

However, in the last ten years, machine learning (ML) and deep learning (DL) methodologies have achieved remarkable success in SRF [9]. For example, some commonly utilized techniques represent for ML can be mentioned as artificial neural networks (ANN) [10], support vector regression (SVR) [11], multilayer perceptron (MLP) [12], and gaussian process regression (GPR) [13] possesses strong nonlinear mapping ability, so it generates advantage in determining the changing trend of solar energy. Sfetsos et al used ANN to make prediction of hourly global radiation and compared them with the conventional procedures based on the clearness index to indicate the more effectiveness of artificial intelligent models [14]. Mellit et al developed an MLP neural network for a 24-hour prediction of solar radiation using the present values of the mean daily solar irradiance and air temperature and compared with the Grid Connected Photovoltaic Plants (GCPV) to show the goodness of this model [12].

In addition, DL approaches were also presented great advantages with better accuracy in SFR such as backpropagation neural network (BPNN) [15], convolutional neural network (CNN) [16]. Feng et al proposed a 21-layer

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deep CNN to forecast 10-min ahead global horizontal irradiance with using only one total sky image as input without any numerical measurements and extra feature engineering and it was concluded that for different weather conditions, CNN outperformed the machine learning model by 11.8% [17]. Besides, recurrent neural network (RNN) [18] was commonly proposed to forecast solar radiation. However, RNN has exploding problems and gradient vanishing so long short-term memory (LSTM) - an improvement of RNN was created to overcome the disadvantages of its predecessor [19]. Therefore, LSTM can handle the dependence on solar energy data for many hours continuously and even long-term information to be collected. Mishra et al proposed a unified architecture for multi-time-horizon predictions for short and long-term solar forecasting using RNN and demonstrated the more effectiveness of this model with random forests, support vector machine, gradient boosting and vanilla feed-forward model [20]. Yu et al applied the LSTM approach based on a timescale to forecast hourly GHI then compared it with ANN and SVR to show the performance of the proposed model [21].

Although the above methods can achieve higher accuracy than the physical and statistical methods, they still have some limitations such as requiring a very large amount of data to perform better than other techniques. Therefore, some hybrid models were created to be able to limit the above disadvantages. The idea of model matching is to use the unique feature of each model to optimize different patterns in the data. That can be an effective way to improve performance forecasting. And some decomposition techniques combined with AI models are the well-known hybrid models in solar radiation prediction, especially empirical mode decomposition (EMD) [22].

In this paper, we proposed a hybrid model based on ensemble empirical mode decomposition (EEMD) and CNN-LSTM for short-term forecasting solar radiation. We decomposed the original data using EEMD into intrinsic mode functions (IMFs) then the CNN-LSTM was used for forecasting each IMFs and finally these IMFs predicted were added for the output results. The proposed method was applied for forecasting solar irradiation collected in Hanoi, Vietnam.

The rest of this paper is organized as follows: Section 2 introduces the methodology consisting of the decomposition technique EEMD and the forecasting model CNN-LSTM. Section 3 provides the dataset, the forecasting results, comparison with other methods and discussion, while section 4 presents the conclusion of the paper.

II. METHODOLOGY

A. Ensemble Empirical Mode Decomposition (EEMD)

EEMD is a new and improved noise-assisted data analysis method of EMD [23]. Its core function is to contain Gaussian white noise into the signal for multiple EMD decays which are its main function, and the result of multiple splits is averaged to obtain the final decomposition result. For the EMD approach, the data $X(t)$ is decomposed into terms of Intrinsic Mode Functions (IMFs), and IMF features are linear and stationary. The EMD method is shown as (1)

$$X(t) = \sum_{j=1}^{n} C_j(n) + R_n \quad (1)$$

Where $C_j(n)$ is the ith IMF component, R_n is the residual of the data after n number of IMFs were decomposed.

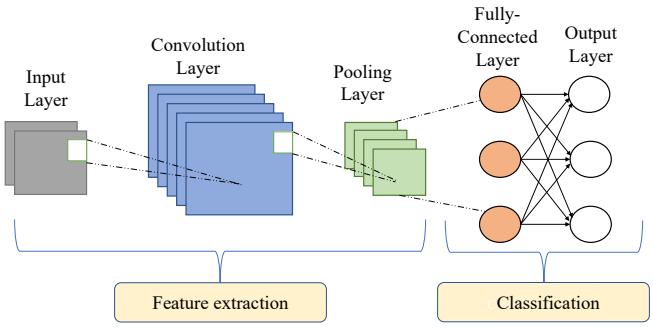


Fig.1. The structure of CNN

However, the EMD approach can not accurately reflect the characteristics of the raw data. And to overcome this shortcoming, EEMD [24] was proposed to calculate the IMF factors by mean of a group of trials, while adding white noise of finite amplitude in each trial. The EEMD performs the following steps:

1. Input the data to be decomposed as the original signal $x(t)$ and add white noise $(\omega_i(t))$ to this input signal to create noise added input $x_i(t)$:

$$x_i(t) = x(t) + \omega_i(t) \quad (2)$$

2. Using the equation (1) to divide $x_i(t)$ into different independent IMFs
3. Repeat steps 1 and 2 using different noises each time
4. IMFs to be averaged to obtain the final result.

B. Convolutional neural network (CNN)

CNN network [17] is a collection of convolutional layers that overlap and use nonlinear activation functions to activate the neuron in the nodes. Its functionality is mainly based on three main functions, the convolution layers, the pooling layers, and the fully connected layers. The architecture of CNN is depicted on **Fig.1**. First, the input data is fed to the input layer. Then in the convolution layer and the pooling layer, the feature extraction is done. Next, this extracted information is assimilated by fully connected layers. Finally, results are shown in the output layer.

In this structure, the convolution layer plays a very important role. The convolution kernels are what is contained in each convolution layer and are calculated by (3). After the convolution layers are active, the data features are extracted, and a composite layer is added after the convolution to reduce the size of the feature.

$$l_t = \tanh(x_t * k_t + b_t) \quad (3)$$

Where l_t denotes the output value after convolution, x_t is the input vector, k_t is the weight of convolution kernel, b_t is the bias of convolution kernel, and tanh is the activation function.

C. Long short-term memory network (LSTM)

The LSTM, proposed by Hochreiter and Schmidhuber, is an improved model of RNN and can deal with vanishing and

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exploding gradients [19]. The LSTM model contains three gates (input, output, and forget) that control what information goes into or out of a memory cell for any arbitrary time. Using the diagram in **Fig.2**, the detailed architecture of a particular memory cell of the LSTM is shown.

Assuming we have the current input x_t (the pre-processed data, which is fed as input data), and output y_t(the prediction output) at time t.

First, the forget gate f_t determines what information is retained, specifying a value between 0 and 1. Values close to 0 are discarded, while values close to 1 are kept.

$$f_t = \sigma.(W_f.[h_{t-1}, x_t] + b_f) \quad (4)$$

New information is added to the current cell state based on the new candidate values thanks to the input gate i_t.

$$i_t = \sigma.(W_i.[h_{t-1}, x_t] + b_i) \quad (5)$$

The update gate u_t generates a vector to be added to the memory cell based on the updated signal by the input gate i_t.

$$u_t = tanh.(W_c.[h_{t-1}, x_t] + b_c) \quad (6)$$

Next, the status of C_{t-1} is upgraded to the new state C_t by removing some unimportant information and adding valuable information

$$C_t = f_t * C_{t-1} + i_t * u_t \quad (7)$$

After that, the hidden layer state h_t of the memory cell and the output gate o_t can be written as (8), (9), respectively.

$$h_t = o_t.\tanh(C_t) \quad (8)$$

$$o_t = \sigma.(W_o.[h_{t-1}, x_t] + b_o) \quad (9)$$

Finally, y_t is the final forecasting output result.

$$y_t = W_y.h_t + b_y \quad (10)$$

In the above formulas, f_t, i_t , u_t , o_t are the output values of the forget gate, input gate, update signal, and output gate, respectively. $W_{i,f,o,c}$ are the weight matrix of each layer. $b_{f,i,o}$ are the bias of each gate and b_y is the output bias. σ is a nonlinear activation function.

D. The proposed model

The proposed model EEMD-CNN-LSTM structure is shown in **Fig.3**. First, the original data was pre-processed by using the EEMD method to separate into sine-form components. At this stage, IMFs with high correlation would be kept making forecasts and IMFs with low correlation would be considered the noise and they had to be removed. After that, we divided each set of IMFs into 2 datasets: training set and testing set with the ratio of 80% and 20% respectively and became the input data for the CNN-LSTM to train and make forecast for each IMFs. Finally, the IMFs were aggregated into the forecast results.

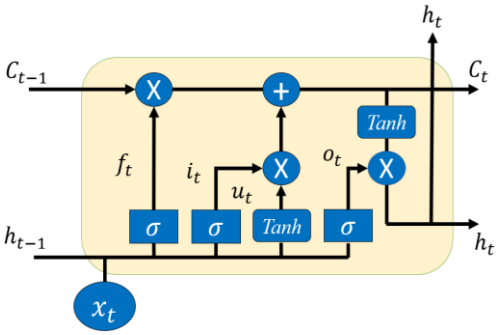


Fig.2. LSTM model functions

E. Performance metrics

A number of metrics for performance were mentioned in this literature [25]. The Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Normalized Root Mean Squared Error are the types of error in this paper, respectively.

Mean Absolute Error (MAE):

$$\text{MAE} = \frac{\sum_{t=1}^{n} |Y_t - \hat{Y}_t|}{n}$$

Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{\sum_{t=1}^{n} (Y_t - \hat{Y}_t)^2}{n}}$$

Normalized Root Mean Squared Error (n-RMSE): The n-RMSE is the normalized form of the RMSE. n-RMSE is a commonly used metric for evaluating solar forecasting accuracy and the error n-RMSE is less than 10%, the forecast result is considered excellent.

$$\text{n-RMSE} = \frac{RMSE}{\bar{y}}$$

Where n is the number of fitted points, Y_t is the actual value, $\hat{Y}_t$ is the predicted value, $\bar{y}$ is the mean of the actual value, Σ is summation notation.

III. Results and Discussion

A. Data collection

The original dataset is the solar radiation in Hanoi city in 2018 and is shown in **Fig.4**. The total number of data points of hourly collected solar radiation is 8760 which is equivalent to one year. The solar radiation data is from National Centre for Hydro - Meteorological Forecasting (NCHMF) and is measured in W/m^2.

B. Data pre-processing

It is unnecessary to deal with missing data in our collected data because the amount of missing data was very small and did not affect the forecast results much. By using the EEMD method, we decomposed data as shown in **Fig.5** and **Fig.6. Fig.6** provided the correlation of the IMF corresponding to **Fig.5**. It can be seen that IMF 1 had a very low correlation so

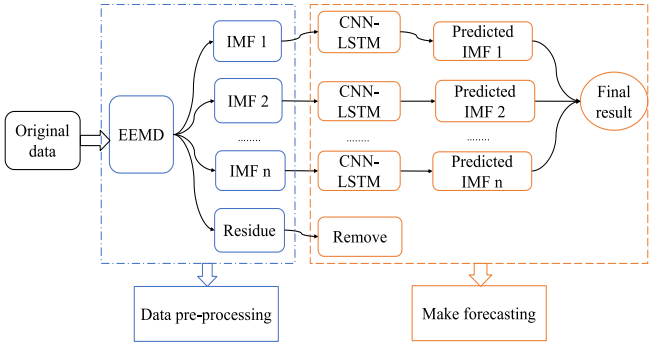


Fig.3. The architecture of proposed model

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it can be considered as white noise while the rest of the IMFs correlated more with its past self. Therefore, the removal of IMF 1 would help improve the forecast results.

C. Forecasting results

After the data were decomposed and denoised, each IMF was fed into the CNN-LSTM model to forecast. The results of each IMF were shown in **Fig.7**. As can be seen from this figure, with the blue line being the actual data and the forecast line being the red dashed line, CNN-LSTM shown its very good forecasting ability. For the very highly correlated IMFs, the prediction shown off a great performance, while for the less correlated IMFs, the learning was still good with negligible deviations.

After forecasting and evaluating the results of each series, these results were aggregated into the final results in **Fig.8** and **Table 1**. In order to illustrate the effectiveness of the proposed method, solar radiation was forecasted using different methods and compare the performance. We applied the single model of CNN [17], LSTM [19] and an improved LSTM which is Bidirectional LSTM (Bi-LSTM) [26] as comparative models. All these models were built again and run with the same cleaned input data as the proposed model. **Fig.8** provided an insight into the performance of all methods are discussed. It is clearly that the red line (proposed model) is closest to the black line (actual) followed by the yellow (Bi-LSTM), orange (CNN) and finally gray line (LSTM).

In **Table 1**, we used 3 error quantities of RMSE, MAE and n-RMSE calculated on testing set to compare the effectiveness of the models' performance. It can be easily analyzed:

- The CNN and Bi-LSTM model performed slightly better than LSTM, but the accuracy was still low with n-RMSE of greater than 15%. These models should be improved or combined with other preprocessing model, decomposition techniques to obtain higher performance in forecasting.
- The proposed hybrid model outperformed other comparative single model concerning all performance matrices with *RMSE*: 19.49, *MAE*: 12.734, *n-RMSE*: 0.098. As mentioned above, with the error of n-RMSE less

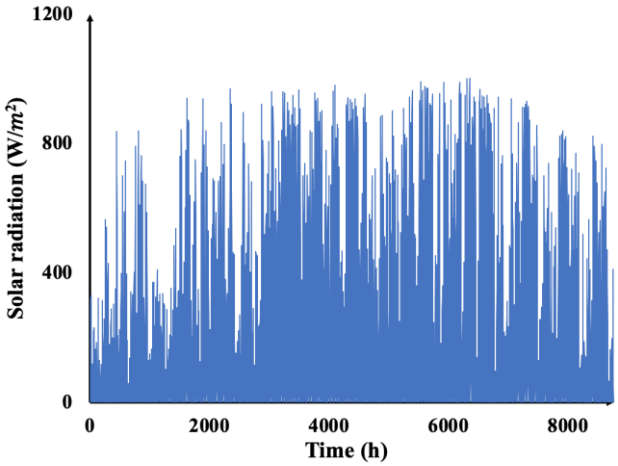


Fig.4: Solar radiation dataset in Hanoi in 2018

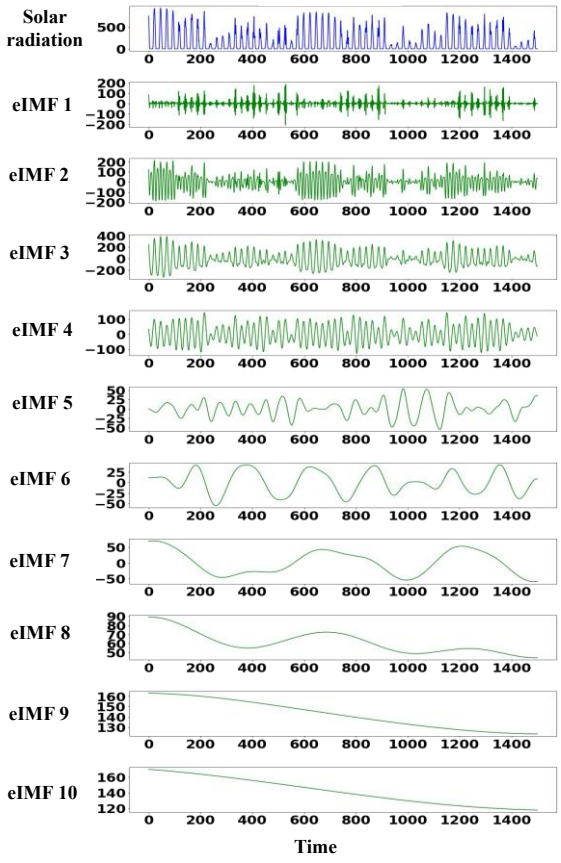


Fig.5: The forecasting of each IMF using CNN-LSTM

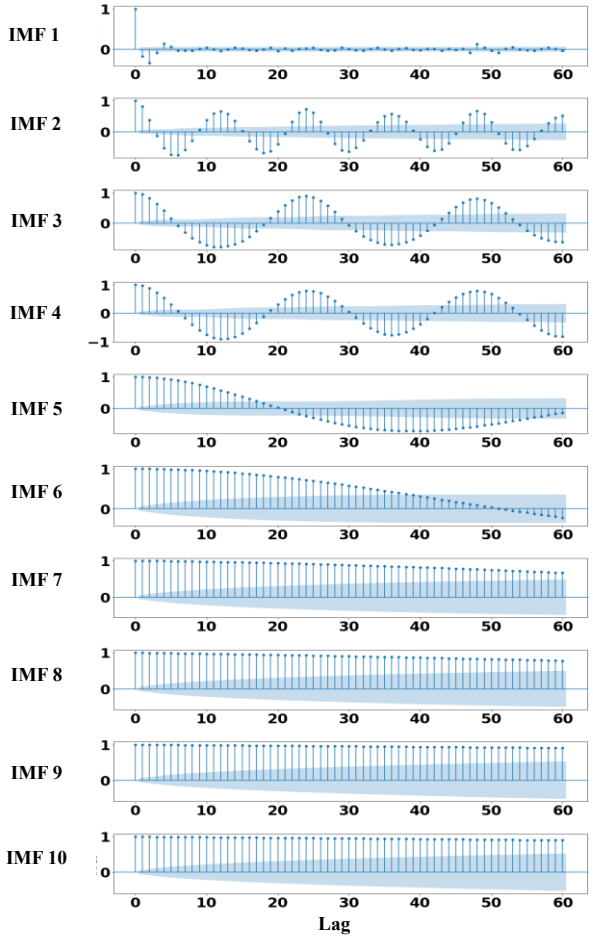


Fig.6: The forecasting of each IMF using CNN-LSTM

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TABLE I. THE PERFORMANCE OF ALL MODELS USING TESTING SET

	Bi-LSTM	CNN	LSTM	Proposed model
RMSE	33.573	35.177	37.186	19.490
MAE	18.054	18.501	18.035	12.734
n-RMSE	0.169	0.177	0.187	0.098

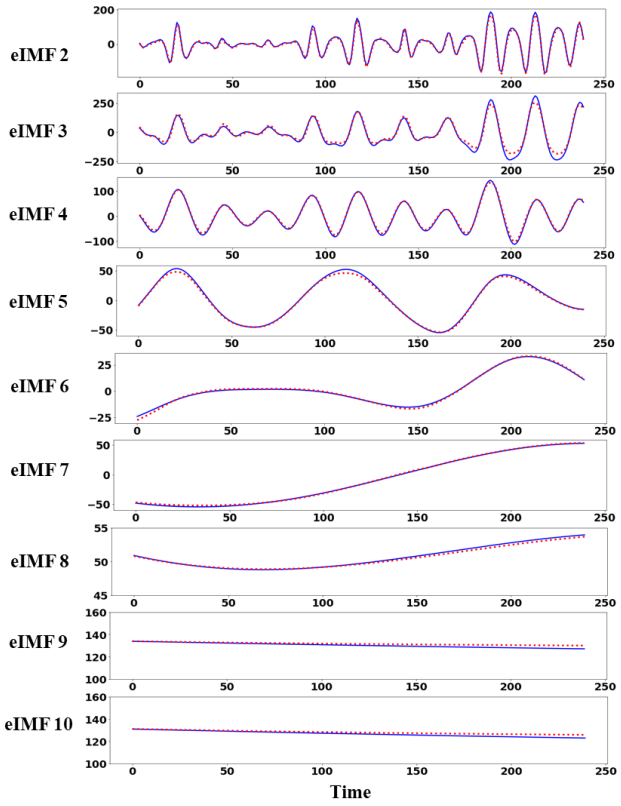


Fig.7: The forecasting of each IMF using CNN-LSTM

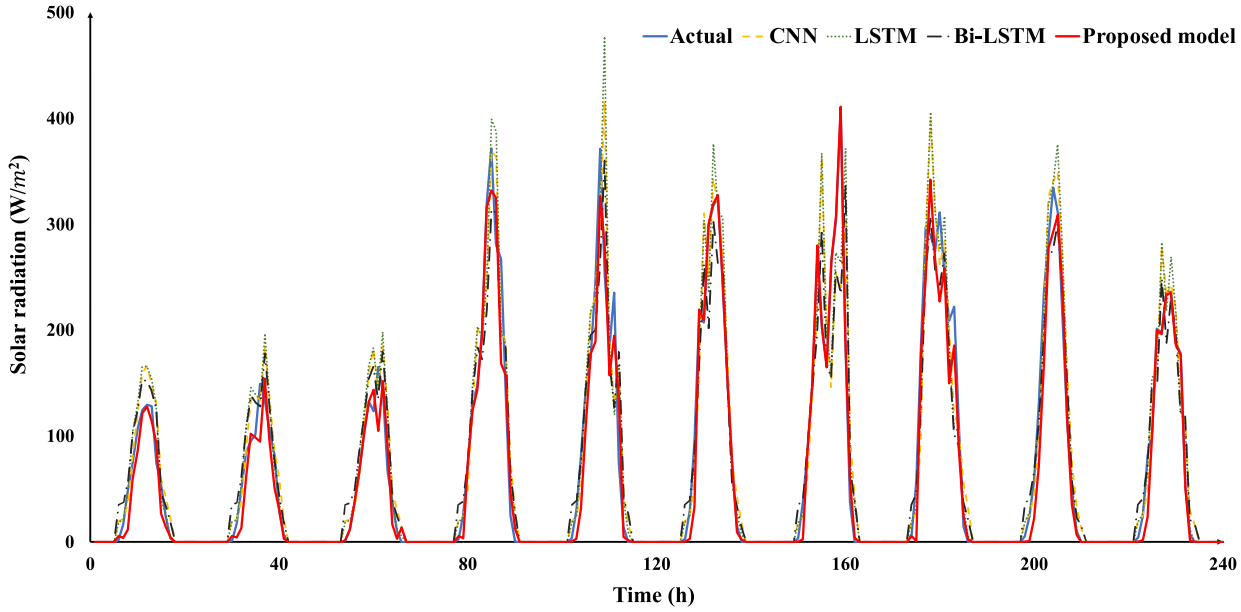


Fig.8: Comparison of forecasting results of the models

than 10% rated as excellent, the proposed model with efficiency of 9.8% was rated as having excellent performance.

IV. CONCLUSION

In this paper, the CNN-LSTM model combined with the EEMD method was proposed to predict solar radiation in Hanoi city in 2018. First, the original data was decomposed to IMFs by using the EEMD method and the residue was removed. After that, the sequences decomposed by EEMD were forecasted using the CNN-LSTM model and finally integrated to obtain the final forecasting result. We used some reference methods such as the LSTM model, CNN model, and Bi-LSTM model to compare and show the performance of the proposed model. From the above results, the proposed model has the highest accuracy with all smallest evaluation metrics, namely *RMSE* of 19.490, *MAE* of 12.734 and *n-RMSE* of 0.098 in which *RMSE* and *n-RMSE* were only half of that of the other comparative models.

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